

National University of Computer and Emerging Sciences

Fast School of Computing

Spring 2024

CS-1004 OOP (CS)

Part A

Quiz # 3 [15 marks]

Time = 15 minutes

Name: _____

Roll No: _____

For the Given code, provide the output:

```
#include <iostream>
using namespace std;
class Num{
    int n;
    static int c;
public:
    Num(){ this->n=4; }
    Num(int nn){ n=nn;c++; display();}
    int getN()const{return this->n;}
    void display(){ n+=1;cout<<n<<" "<<c<<endl; }
    ~Num(){ cout<<c<<" "<<n<<endl; --c; }
}nn,*ptr;
int Num::c=0;
int main() {
    int n=8;
    if(true){
        ptr =new Num(nn.getN());
        const Num c(n);
    }
    nn.display();
    cout<<"-----"<<endl;
    return 0;
}
```

output:

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Part B

Quiz # 3 [10 marks]

Time = 15 minutes

Name: _____

Roll No: _____

For the Given code, provide the output:

```
#include <iostream>
using namespace std;
class Num{
    int n;
    static int c;
    public:
    Num(){ this->n=7; }
    Num(int nn){ n=nn;c++; display();}
    int getN()const{return this->n;}
    void display(){ n+=c;cout<<n<<" "<<c<<endl; }
    ~Num(){ cout<<c<<" "<<n<<endl; --c; }
}nn,*ptr;
int Num::c=0;
int main() {
    int n=8;
    if(true){
        ptr =new Num(nn.getN());
        const Num c(n);
    }
    nn.display();
    cout<<"-----"<<endl;
    return 0;
}
```

output: